Flood Risk (Depth-Damage) — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 27-March-2026

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1 Test Results — Flood Risk (Depth-Damage) (MKM-DD-001)

Tests run on **27 March 2026 at 11:52:59**.

Table 1: Test Results Summary — Flood Risk (Depth-Damage) (MKM-DD-001)

Total Tests	32
Passed	32
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Flood Risk (Depth-Damage)

Test Description	Acceptance Criteria	Result	Time
Every hour from 0..n must be present in output, in order.	`hours == list(range(len(gauge_readings)))`	Pass	0.000s
Flood depth must be non-negative at every time step.	`r['depth_m'] >= 0.0`	Pass	0.000s
Each hydrograph entry must contain hour, wse_m, depth_m, and flooded.	`'hour' in r; 'wse_m' in r` (+2 more)	Pass	0.000s
Empty gauge readings must return empty property hydrograph.	`build_property_hydrograph([], 8.0, 1.0, 0.9, 6.0, 0.3) == []`	Pass	0.000s
When floor level exceeds peak WSE, no time step should be flooded.	`r['flooded'] is False`	Pass	0.000s
With travel time of 5 hours, first 5 hours must show zero flood depth.	`r['depth_m'] == 0.0; r['flooded'] is False`	Pass	0.000s
Property hydrograph must have same number of time steps as gauge input.	`len(result) == len(gauge_readings)`	Pass	0.000s
Low-elevation property at gauge location must flood at peak water level.	`max(peak_depths) > 0`	Pass	0.000s
Slope from 10m to 8m over 1000m should be 0.002 (2m/1000m).	`abs(s - 0.002) < 1e-6`	Pass	0.000s
Flat terrain (equal elevations) must return MIN_SLOPE, not zero.	`s == MIN_SLOPE`	Pass	0.000s
Slope must be the same regardless of flow direction (uses absolute value).	`s1 == s2`	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Zero horizontal distance must return MIN_SLOPE to avoid division by zero.	t == 0.0; s == MIN_SLOPE	Pass	0.000s
Monotonicity: deeper water must produce higher velocity at constant slope.	v2 > v1	Pass	0.000s
Negative depth is physically invalid; must return zero velocity.	compute_manning_velocity(-1.0, 0.005) == 0.0	Pass	0.000s
Negative slope (downhill in opposite direction) must use absolute value.	v_pos == v_neg	Pass	0.000s
Manning velocity at d=1.0m, S=0.005, n=0.04 should be ˜1.77 m/s.	1.5 < v < 2.0	Pass	0.000s
Slopes below MIN_SLOPE must be clamped to MIN_SLOPE to avoid	v == v_min	Pass	0.000s
Monotonicity: steeper slope must produce higher velocity at constant depth.	v2 > v1	Pass	0.000s
Zero depth must produce zero velocity (no flow without water).	compute_manning_velocity(0.0, 0.005) == 0.0	Pass	0.000s
Zero roughness coefficient causes division by zero; must return 0.	compute_manning_velocity(1.0, 0.005, roughness=0.0) == 0.0	Pass	0.000s
At distance = characteristic length, factor must equal 1/e 0.368.	abs(f - 1.0 / math.e) < 0.001	Pass	0.000s
compute_attenuation is a backwards-compatible alias.	compute_attenuation(800) == compute_retention(800)	Pass	0.000s
At 5x the decay length, retention should be ¡ 1%.	f < 0.01	Pass	0.000s
At 800m with default 10km length scale, retention ¿= 0.92.	f >= 0.92	Pass	0.000s
Negative distance is clamped to zero; must return 1.0.	compute_retention(-100) == 1.0	Pass	0.000s
At 1km with default 10km length scale, retention ¿= 0.90.	f >= 0.90	Pass	0.000s
At zero distance, retention must be 1.0 (full signal).	compute_retention(0) == 1.0	Pass	0.000s
Zero decay length means instant decay; must return 0.0.	compute_retention(100, 0) == 0.0	Pass	0.000s
Travel time for 500m at d=1.0m, S=0.005 should be well under 1 hour.	0 < t < 1.0	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Monotonicity: greater distance must produce longer travel time.	`t2 > t1`	Pass	0.000s
Zero depth produces zero velocity, so travel time must be infinite.	`t == float('inf')`	Pass	0.000s
Zero distance must produce zero travel time.	`t == 0.0; s == MIN_SLOPE`	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (29 tests)	D. Kelly
07-Mar-2026	1.1	29 test(s) added; 29 test(s) removed (total: 29)	D. Kelly
18-Mar-2026	1.2	117 test(s) added (total: 146)	D. Kelly
22-Mar-2026	1.3	18 test(s) removed (total: 128)	D. Kelly
23-Mar-2026	1.4	99 test(s) removed (total: 29)	D. Kelly
27-Mar-2026	1.5	8 test(s) added; 5 test(s) removed (total: 32)	D. Kelly